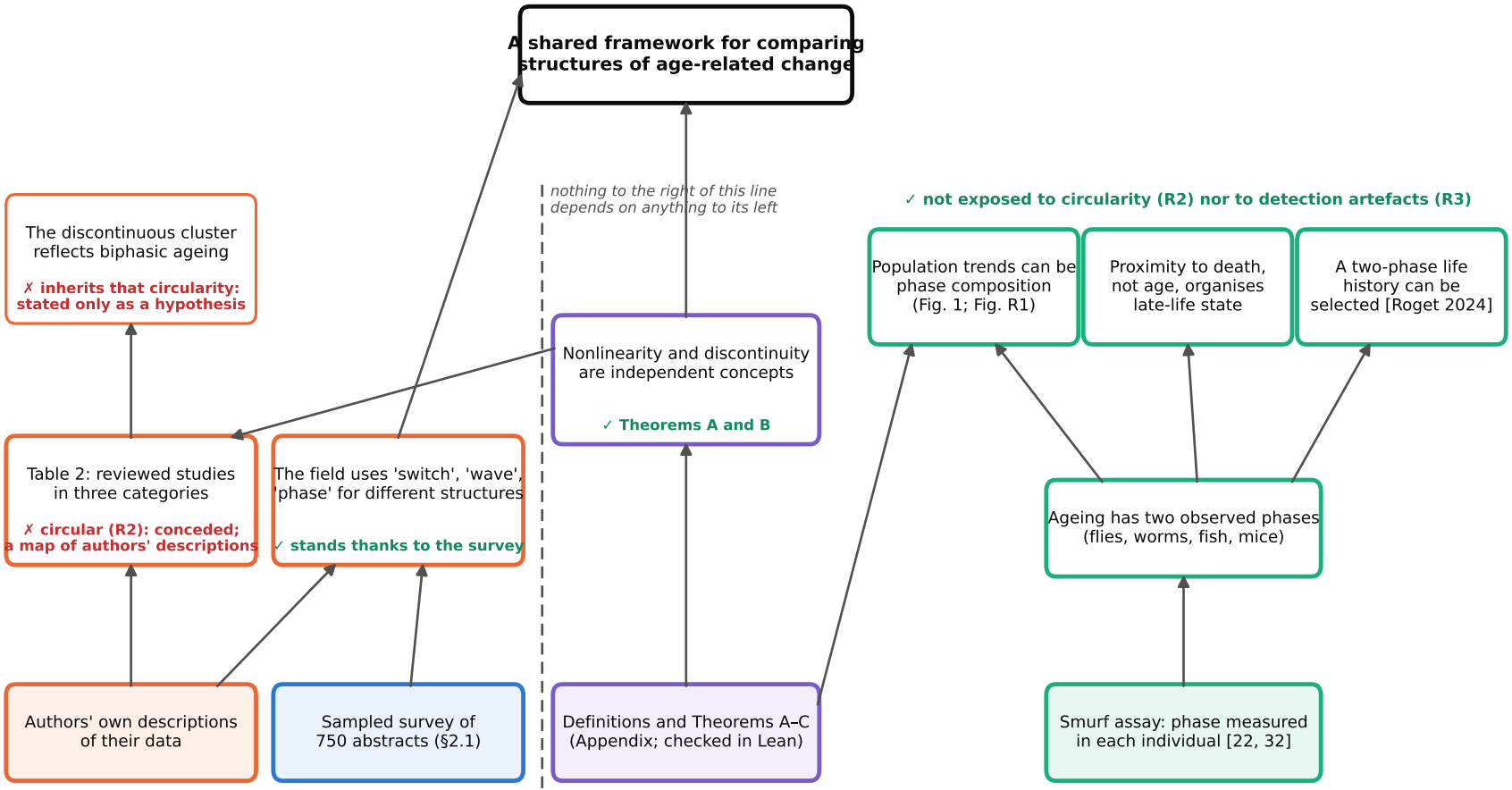
The literature map (conceptual)
The two-phase model (empirical)
A shared framework for comparing structures of age-related change
nothing to the right of this line depends on anything to its left
The discontinuous cluster reflects biphasic ageing
✗ inherits that circularity: stated only as a hypothesis
Nonlinearity and discontinuity are independent concepts
✓ Theorems A and B
✓ not exposed to circularity (R2) nor to detection artefacts (R3)
Population trends can be phase composition (Fig. 1; Fig. R1)
Proximity to death, not age, organises late-life state
A two-phase life history can be selected [Roget 2024]
Table 2: reviewed studies in three categories
✗ circular (R2): conceded; a map of authors' descriptions
The field uses 'switch', 'wave', 'phase' for different structures
✓ stands thanks to the survey
Ageing has two observed phases (flies, worms, fish, mice)
Authors' own descriptions of their data
Sampled survey of 750 abstracts (§2.1)
Definitions and Theorems A–C (Appendix; checked in Lean)
Smurf assay: phase measured in each individual [22, 32]